

AI-Based Student Dropout Early Warning: A Leakage-Aware Landmark Evaluation on UCI and OULAD

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Abstract

Background. Dropout prediction can appear more accurate when models use information recorded after withdrawal or when repeated records from one student are divided between training and test data. **Objective.** We evaluated future withdrawal at fixed decision dates and separated the operational effect of a changing risk set from the information gained through additional behavioural history. **Methods.** We used the UCI student dataset (4,424 students) and OULAD (32,593 student-module records; 10,655,280 interaction rows). OULAD day-35, day-60, and day-75 cohorts included only students registered and still enrolled at each cutoff. Students were grouped into mutually exclusive training, validation, and test sets. Logistic regression and XGBoost were evaluated with validation-only tuning and calibration. Uncertainty was estimated with 1,000 student-cluster bootstrap samples. **Results.** In dynamic OULAD cohorts, XGBoost ROC-AUC remained near 0.717, while average precision declined from 0.356 at day 35 to 0.277 at day 75. In a controlled day-75 cohort with the same records and target at each window, ROC-AUC rose from 0.673 to 0.718 (paired difference 0.045; 95% CI 0.030 to 0.061). ROC-AUC was 0.678 on a later 2014J presentation holdout. **Conclusions.** Additional history improved ranking for fixed cases, but later operational cohorts were smaller and harder. The model may support a human-reviewed advising queue, but it requires local prospective validation before institutional use.

Keywords: student dropout; early warning; educational data mining; landmark prediction; OULAD; calibration; SHAP

1. Introduction

1.1 Business and Educational Context

A student can show several warning signs before leaving a course: missed assessments, weak scores, late submissions, or a decline in online activity. Advisors may see parts of this information, but it is usually distributed across systems and arrives at different times. The practical question is therefore not only whether a student will eventually withdraw. It is whether the institution can identify a useful group for support while there is still time to respond.

This distinction matters because a model may classify final outcomes accurately by using first- or second-semester grades, yet provide little warning before the outcome. A deployable model also needs probabilities that reflect observed risk, a manageable alert volume, and an evaluation that prevents the same student from appearing in both training and test data.

1.2 Research Questions

- RQ1. How closely can the main UCI multiclass experiment from Islam et al. [1] be reproduced with the available data and documented model family?
- RQ2. How accurately can future OULAD withdrawal be predicted at days 35, 60, and 75 among students who are still enrolled at each decision date?

- RQ3. When the students and outcome are fixed, does adding behavioural history from day 35 to day 75 improve ranking?
- RQ4. How do performance, calibration, and feature importance change across time and across a later course-presentation holdout?
- RQ5. What operational and ethical conditions are required before using the model as an advisor decision aid?

1.3 Contributions

Our original milestones compared common classifiers, tuned selected models, and used SHAP. Those components alone were not a strong research contribution because similar work already exists. The final study adds methodological depth in four places. First, it uses outcome timing to construct landmark risk sets. Second, it creates student-level splits with no cross-set overlap. Third, it reports both a deployable dynamic-cohort analysis and a controlled same-record analysis. Fourth, it adds calibration, clustered uncertainty intervals, and a later-presentation generalization test.

1.4 Scope and Computing Constraints

The project is a retrospective secondary analysis of two public, de-identified datasets. It does not evaluate an intervention and does not estimate a causal effect of advising. All experiments were completed on a Mac laptop. This supported interpretable tabular models and 1,000 cluster-bootstrap resamples, but not a large deep-learning search. We treated this as a useful constraint: the selected models are easier to reproduce and more realistic for an institutional pilot.

2. Related Work and Research Position

2.1 Base Paper and UCI Benchmark

Islam et al. [1] used the UCI student dataset and reported 83% accuracy for XGBoost in a three-class academic-performance task. The study also used several explanation tools and proposed a student support system. Realinho et al. [2] describe the same UCI data as 4,424 student records with enrolment-time variables and performance variables measured after the first and second semesters. This timing is important: the full dataset is suitable for final-outcome classification, but it should not automatically be described as day-one prediction.

Our reproduction keeps the multiclass task for comparability. A separate UCI benchmark then removes every first- and second-semester curricular variable. That stricter benchmark asks what can be known from admission and enrolment information only.

2.2 OULAD and Time-Aware Prediction

OULAD combines demographics, registration, assessments, and daily Virtual Learning Environment activity [4]. It contains 22 module-presentations, 32,593 student-module records, and more than 10 million interaction rows. The source paper notes that B and J presentations can differ, so presentation-aware evaluation is important. Recent OULAD studies and public implementations also show that model scores depend strongly on the selected horizon, outcome definition, features, and split design [6, 7]. Direct comparison of headline accuracy values is therefore unsafe unless these choices match.

Two recent studies are especially close to the final version of this project. Wu et al. [6] combined early-window OULAD features with calibration, capacity rules, fairness auditing, and presentation-grouped evaluation. Their target was final-result withdrawal rather than withdrawal occurring after a landmark. Da Silva et al. [16] used valid withdrawal dates in a broader survival benchmark. These studies narrow our novelty claim. Our remaining distinction is the direct side-by-side comparison of a changing operational risk set and a fixed same-record cohort.

2.3 Explainability and Reporting Quality

SHAP can summarize how a fitted model uses its inputs [9]. It does not show that changing a feature will cause a student to persist. For example, assessment score and submission timing may reflect several underlying conditions.

We use SHAP as a descriptive model audit and group related inputs into academic, engagement, program, demographic, and prior-preparation families.

There is no single required reporting guideline for this capstone. We adapted the transparency principles in TRIPOD+AI [10] and the risk-of-bias questions in PROBAST+AI [11]. These guidelines were developed for prediction-model research more broadly and are not claimed as educational validation standards. They helped us report the data source, eligibility rules, prediction time, outcome, splits, tuning, calibration, uncertainty, and intended use more clearly.

Table 1. Research gap addressed by the final study

Common approach	Risk	Our response
Random row split	The same student may appear in training and test data	Group split by student identifier
Final withdrawal label at an early cutoff	Students already withdrawn may be counted as predicted	Landmark risk set and future-withdrawal outcome
One temporal result	Cohort change is mixed with information gain	Dynamic and fixed-cohort analyses
AUC or F1 only	Probability quality and uncertainty are hidden	AP, Brier score, calibration, and clustered CIs
Random holdout only	Later-period generalization is unknown	2014J presentation holdout

3. Materials and Methods

3.1 Study Design and Data Sources

Table 2. Data sources and analytical roles

Dataset	Size	Main variables	Role
UCI	4,424 students; 36 predictors	Admission, demographics, finance, semester records	Base-paper reproduction and enrolment-time benchmark
OULAD	32,593 records; 10,655,280 VLE rows	Registration, demographics, assessment, daily VLE activity	Primary landmark and temporal evaluation

The UCI target has three outcomes: Dropout (1,421), Enrolled (794), and Graduate (2,209). For the enrolment-time binary task, Dropout is positive and the other two classes are grouped as non-dropout. This grouping leaves an unresolved Enrolled category inside the negative class and is a source of label noise.

OULAD records a student's result for a module-presentation: Withdrawn, Fail, Pass, or Distinction. The prediction unit in this study is the student-module-presentation record. A student can appear in more than one record, which is why splitting by row is not sufficient.

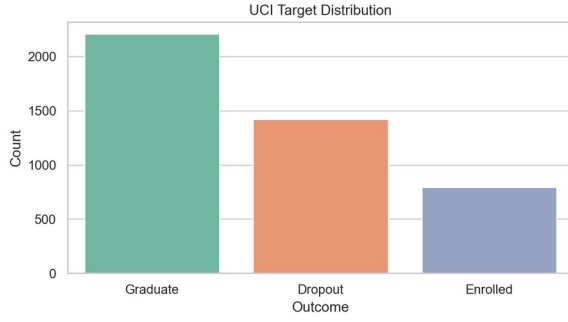


Figure 1a. UCI outcome distribution.

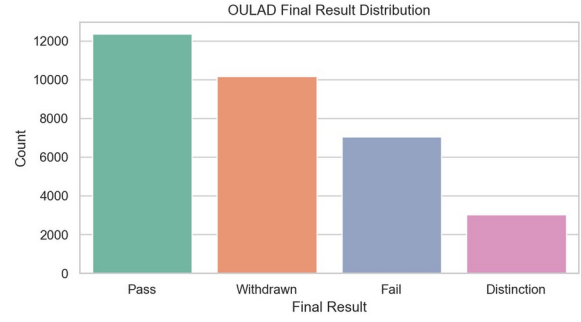


Figure 1b. OULAD final-result distribution.

3.2 Landmark Cohort and Outcome Definition

For each OULAD cutoff t in $\{35, 60, 75\}$, a record was eligible if the student had registered on or before t and had no recorded unregistration on or before t . The binary outcome was 1 only when the final result was Withdrawn and the recorded unregistration date was after t . Withdrawn records without a known unregistration date were excluded because their temporal order could not be confirmed.

This is a landmark prediction design. The day-35 model therefore answers: among students still enrolled on day 35, who later withdraws? It does not receive credit for identifying students who had already left before day 35.

Table 3. OULAD landmark cohorts

Cohort	Rows	Unique students	Future withdrawals	Rate
Day 35	27,129	24,567	4,718	17.39%
Day 60	26,253	23,899	3,837	14.62%
Day 75	25,795	23,566	3,378	13.10%

3.3 Predictor Construction

At each cutoff, OULAD predictors were rebuilt using only events dated on or before that cutoff. Static predictors included module, presentation, registration date, studied credits, prior attempts, education, age band, gender, region, deprivation band, and disability. Behavioural predictors summarized total and activity-specific clicks, active days, unique resources, recent activity, assessment availability, submission ratio, weighted score, score range, and submission delay.

For the UCI enrolment-time benchmark, all first- and second-semester curricular variables were removed. Predictors included admission grade, previous qualification, application information, demographics, financial indicators, and macroeconomic variables. The full-feature UCI data were used only for the base-paper reproduction.

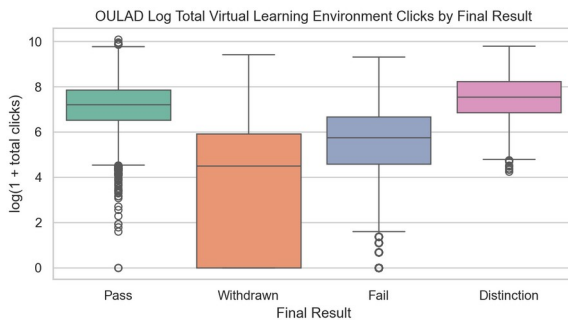


Figure 2a. OULAD click activity by final result.

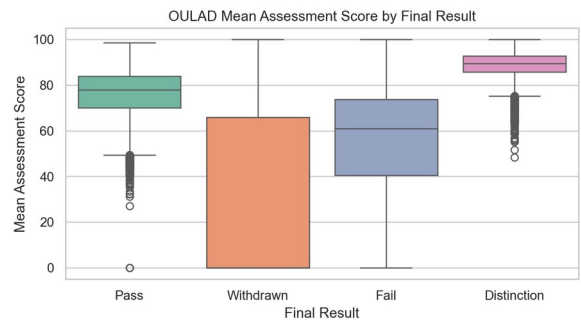


Figure 2b. OULAD assessment score by final result.

3.4 Missing Data and Encoding

Numeric missing values were filled with the training-set median. Categorical missing values were assigned to an explicit most frequent category through the fitted preprocessing pipeline. Categorical predictors were one-hot encoded. All imputation and encoding steps were fit on training data and then applied to validation and test data.

3.5 Split Design

OULAD students were assigned to 60% training, 20% validation, and 20% test groups. The assignment was performed on unique student identifiers, then joined back to every student-module record. Across all three cutoffs, the observed train-validation, train-test, and validation-test student overlap was zero. The same group assignment was reused across windows.

A second split tested later-presentation generalization. Models were trained on 2013B and 2013J, model choices were made on 2014B, and the final test used 2014J. Any student appearing in an earlier partition was removed from the later one. This is a temporal sensitivity analysis within OULAD, not validation at a different institution.

3.6 Models, Tuning, and Calibration

We compared logistic regression with XGBoost. Logistic regression served as the simpler benchmark. XGBoost was retained because it can represent nonlinear effects and interactions in mixed tabular data [8]. Class weighting was based on the training outcome. For the main OULAD analysis, a small grid was evaluated only on the day-35 validation set. The earliest window was used so that model complexity was not selected in favor of the later windows. The selected settings were then held fixed across all three cutoffs.

The search was intentionally limited. It compared model sizes that could be completed and explained on the available laptop. It was not a claim that the global optimum had been found. Validation cost included 12 XGBoost combinations and five logistic values. After fitting the model, a logistic calibration layer was trained on validation predictions. An F1-maximizing decision threshold was also chosen on validation data. Test labels were not used for model, calibration, or threshold selection.

3.7 Evaluation and Statistical Analysis

ROC-AUC measured ranking across all thresholds. Average precision (AP) was included because future withdrawal was the minority outcome and prevalence changed across cutoffs. Precision, recall, F1, specificity, and confusion-matrix counts described the validation-selected operating point. Brier score and calibration intercept and slope assessed probability quality. Brier skill compared the model with a constant training-prevalence forecast.

Uncertainty intervals were estimated with 1,000 bootstrap samples at the student level. All records from a sampled student received the same bootstrap weight. For the controlled temporal comparison, the same bootstrap draw was used for both windows before computing the metric difference. A 95% percentile interval excluding zero was treated as evidence that the window difference was not explained by resampling variation alone.

3.8 Two Temporal Estimands

Table 4. Why two temporal analyses were required

Analysis	What is held fixed?	Question answered	Use
Dynamic landmark	Cutoff definition and split rule	How well does the model work on students still enrolled now?	Operational estimate
Controlled day-75 cohort	Records, students, outcome, settings, and split	Does more history improve ranking for the same cases?	Explanatory comparison

The controlled cohort contains students known, in retrospect, to remain eligible through day 75. It is therefore not a deployable day-35 cohort. Its purpose is to isolate information accumulation from population change.

3.9 Explainability

TreeSHAP values were calculated for the uncalibrated XGBoost ranking model on held-out records [9]. Mean absolute SHAP values were grouped into five feature families. The analysis describes which variables the model used and how that balance changed across windows. It does not establish causal mechanisms or recommend an intervention for a specific student.

4. Results

4.1 Reproduction and Audit of Preliminary Results

The base-paper reproduction used the full UCI predictors and XGBoost for the three-class outcome. Test accuracy was 0.768 and macro F1 was 0.704, compared with the 0.83 accuracy reported by Islam et al. [1]. The exact training protocol and all random seeds from the paper were not available, so this is a close reproduction rather than an exact replication.

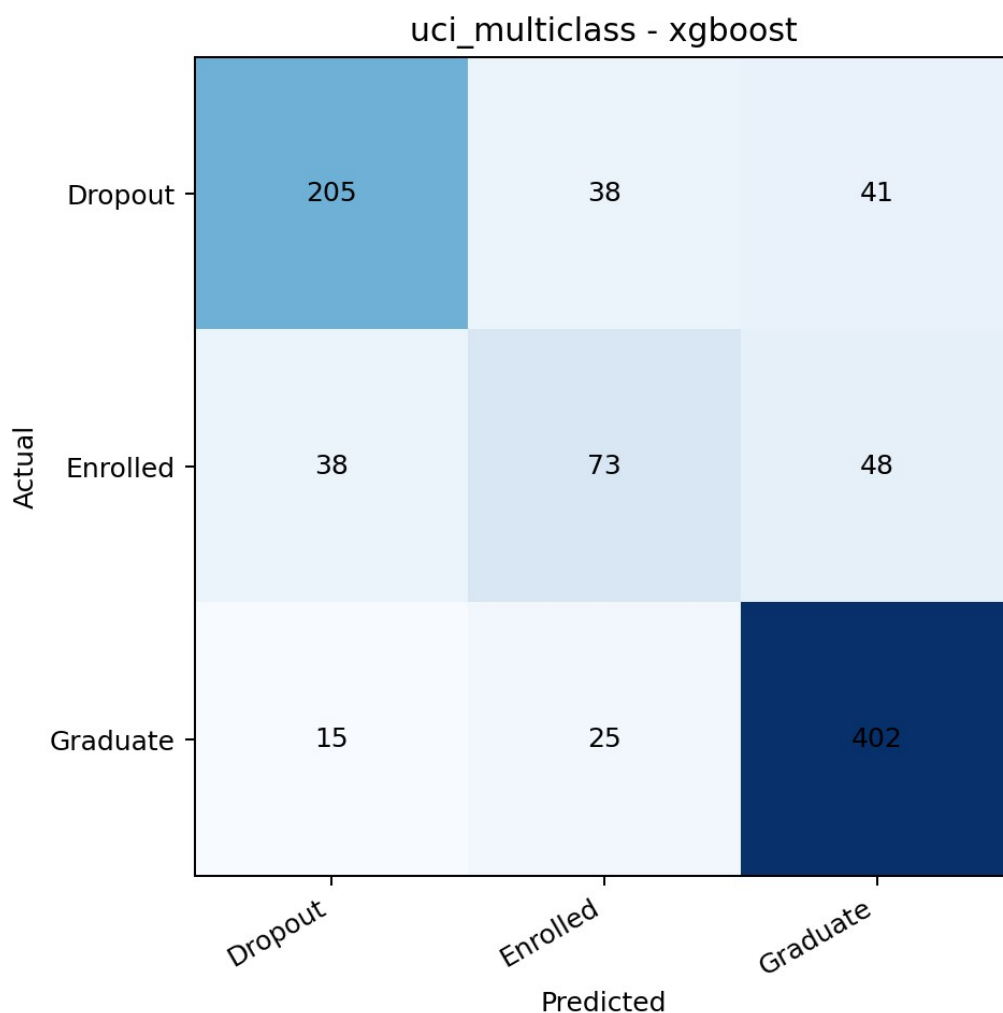


Figure 3. Held-out confusion matrix for the UCI multiclass XGBoost reproduction.

Our milestone models originally reported UCI F1=0.793 and OULAD F1=0.738. Those experiments were useful for learning the pipeline, but they are not the primary results of this report. The UCI model used first-semester records without a known withdrawal date, so it was retrospective classification. The OULAD model used a row-level split and included students who had already withdrawn by the cutoff. The audit found that 17.6% of unique students in the old test records also appeared in training or validation data. These choices made the original result too optimistic.

Table 5. Methodological audit and correction

Issue in preliminary work	Evidence	Final-report correction
OULAD row-level split	Student overlap across partitions	Group-exclusive student split
Already-withdrawn cases at cutoff	52.6% of withdrawals occurred by day 35	Future withdrawal among students still enrolled
UCI called early warning	First-semester data lack withdrawal timing	Separate enrolment-time benchmark
Weighted scores treated as probabilities	Poor preliminary calibration	Validation-only probability calibration
Row bootstrap	Repeated records from some students	Student-cluster bootstrap

4.2 UCI Enrolment-Time Benchmark

Table 6. UCI day-one test performance

Model	Precision	Recall	F1	ROC-AUC	AP	Brier
Logistic regression	0.579	0.732	0.647	0.833	0.743	0.144
XGBoost	0.576	0.789	0.666	0.832	0.749	0.144

The two models had almost identical ROC-AUC and Brier score. XGBoost produced higher recall and F1 at its validation-selected threshold, while logistic regression offered nearly the same ranking with a simpler model. This result suggests that nonlinear complexity added limited value for the UCI enrolment-time inputs.

4.3 Operational OULAD Landmark Performance

Table 7. XGBoost performance in dynamic landmark test cohorts

Cutoff	Test n	Prev.	Precision	Recall	F1 (95% CI)	ROC-AUC (95% CI)	AP (95% CI)
Day 35	5,399	16.84%	0.310	0.575	0.403 (0.379-0.427)	0.717 (0.697-0.735)	0.356 (0.328-0.389)
Day 60	5,219	13.93%	0.299	0.516	0.379 (0.350-0.406)	0.718 (0.696-0.739)	0.305 (0.274-0.338)
Day 75	5,135	12.52%	0.261	0.499	0.343 (0.316-0.372)	0.717 (0.695-0.739)	0.277 (0.247-0.313)

Ranking remained stable near 0.717 ROC-AUC. AP and F1 declined at later cutoffs. This does not mean that later activity had no value. The students who withdrew early were no longer eligible, so the remaining positive cases became less common and more difficult. AP is sensitive to that lower prevalence, and the validation-selected operating point also changed.

4.4 Controlled Same-Record Temporal Comparison

Table 8. Fixed day-75 cohort XGBoost performance

History available	F1	ROC-AUC	AP	Brier score
Through day 35	0.306	0.673	0.236	0.105
Through day 60	0.339	0.700	0.267	0.102
Through day 75	0.343	0.718	0.278	0.101

Table 9. Paired student-cluster bootstrap differences

Comparison	Metric	Mean change	95% CI
Day 60 - day 35	ROC-AUC	+0.027	0.013 to 0.039
Day 60 - day 35	AP	+0.030	0.013 to 0.048
Day 75 - day 60	ROC-AUC	+0.018	0.009 to 0.028
Day 75 - day 60	AP	+0.011	-0.004 to 0.025
Day 75 - day 35	ROC-AUC	+0.045	0.030 to 0.061
Day 75 - day 35	AP	+0.041	0.022 to 0.060

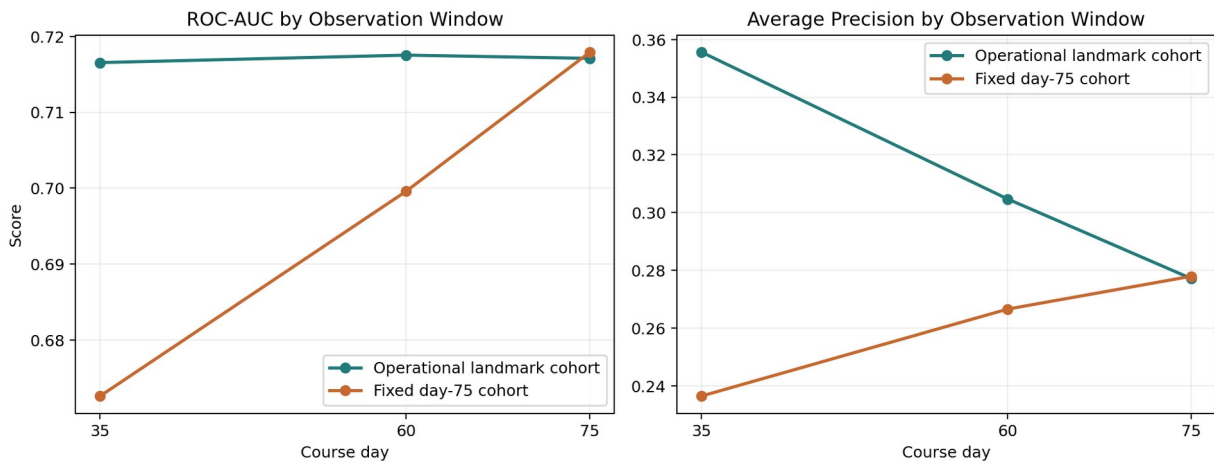


Figure 4. Dynamic landmark performance and the controlled same-record temporal comparison. Error bars are 95% student-cluster bootstrap intervals.

The day-35 to day-60 and day-35 to day-75 intervals excluded zero for both metrics. From day 60 to day 75, the ROC-AUC interval excluded zero but the AP interval did not. The evidence therefore supports an overall gain from day 35 to day 75, with weaker evidence of an additional AP gain after day 60.

4.5 Later-Presentation Holdout

Table 10. Day-75 performance on the 2014J holdout

Model	Test rows	Precision	Recall	F1	ROC-AUC	AP
Logistic regression	8,526	0.252	0.254	0.253	0.609	0.217

Model	Test rows	Precision	Recall	F1	ROC-AUC	AP
XGBoost	8,526	0.225	0.453	0.301	0.678	0.229

XGBoost remained better than logistic regression, but its ROC-AUC fell from 0.717 in the random student holdout to 0.678 in the later presentation. AP was 0.229, compared with a test prevalence of 12.84%. This gap is evidence of temporal or presentation shift and supports periodic local revalidation.

4.6 Calibration

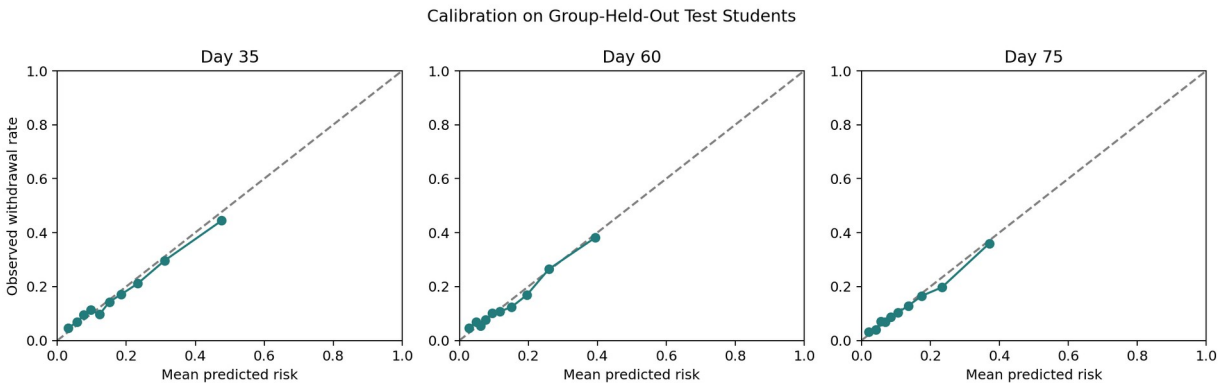


Figure 5. Calibration curves for dynamic OULAD XGBoost models. The diagonal indicates agreement between predicted and observed risk.

After validation-only calibration, Brier scores were 0.126, 0.110, and 0.101 at days 35, 60, and 75. Brier skill scores relative to a constant training-prevalence forecast were 0.097, 0.087, and 0.076. Calibration slopes ranged from 0.88 to 0.91, with negative intercepts from -0.18 to -0.24 . The probabilities were useful but not perfectly calibrated, especially at the high-risk end where observations were sparse.

4.7 Feature Importance Shifts

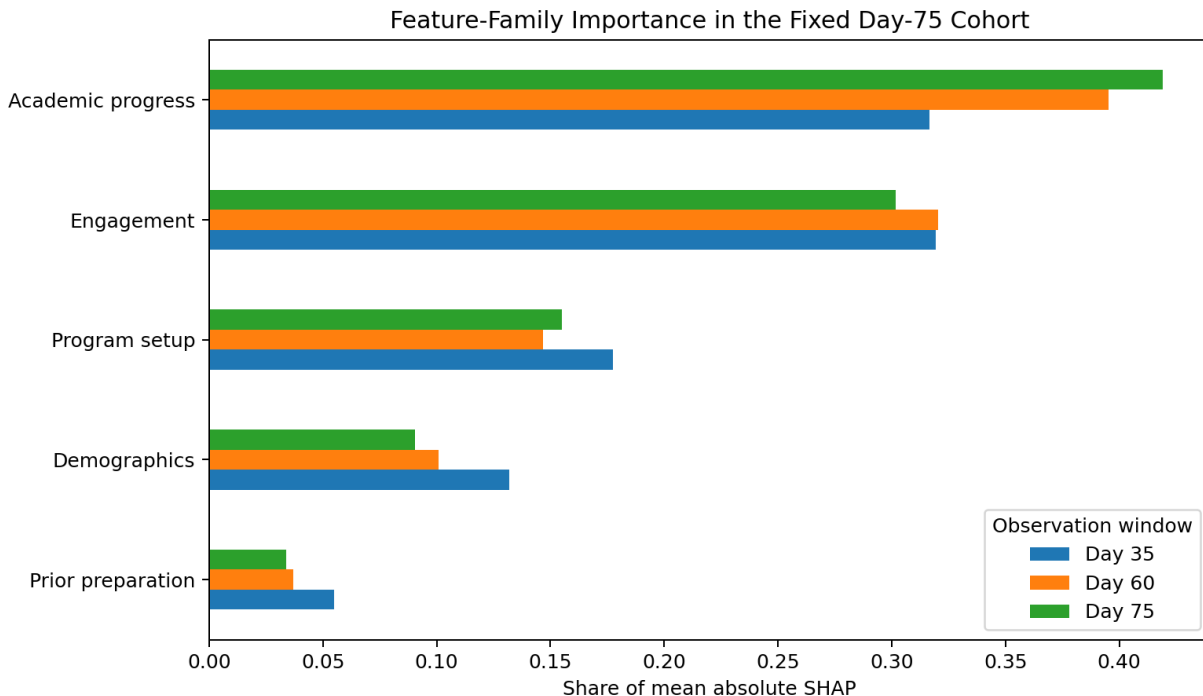


Figure 6. Feature-family share of mean absolute SHAP in the controlled day-75 cohort.

Academic-progress importance increased from 31.7% at day 35 to 39.5% at day 60 and 41.9% at day 75. Engagement remained important and shifted from 31.9% to 30.2%. Prior preparation fell from 5.5% to 3.4%. By day 75, high-ranking variables included weighted assessment score, score relative to available assessment weight, submission delay, expected assessment weight, wiki clicks, minimum score, forum activity, mean score, studied credits, registration date, and active days.

The shift is consistent with a reasonable educational pattern: early predictions depend more on broad participation and context, while later predictions can use direct evidence of academic progress. This remains an association inside the model. It does not show that changing a click count or score alone would prevent withdrawal.

4.8 Capacity-Based Operating Scenario

Table 11. Illustrative 15% validation-capacity threshold

Cutoff	Test alerts	Precision	Recall	F1	True alerts	False alerts	Missed
Day 35	820	0.401	0.362	0.381	329	491	580
Day 60	797	0.349	0.382	0.365	278	519	449
Day 75	764	0.305	0.362	0.331	233	531	410

The threshold was set at the 85th percentile of validation risk scores, representing a service that plans to review roughly the highest-risk 15% of cases. The test alert rate differs slightly because the score distribution changed. At day 35, the model found 329 of 909 future withdrawals and generated 491 false alerts. These counts show why the score should prioritize a review queue rather than automatically trigger a high-stakes action.

5. Discussion

5.1 Principal Findings

The main result is not a single best F1 score. It is the difference between two valid but distinct temporal questions. In a fixed cohort, adding information through day 60 and day 75 improved ranking with paired intervals that excluded zero. In an operational cohort, performance did not rise because many early-withdrawing students had already left and the later risk set contained fewer, harder cases. A useful early-warning study must state which of these questions it is answering.

The day-35 operational model had moderate discrimination (ROC-AUC 0.717) and AP that was about twice its positive rate. It was not accurate enough to justify autonomous decisions. The later-presentation result was weaker, which further limits a deployment claim. At the same time, calibration added value beyond the class-weighted raw scores, and the student-level intervals showed that the fixed-cohort temporal gains were not only the result of one random sample.

5.2 Comparison with Prior Work

The 0.768 UCI reproduction accuracy was below the 0.83 reported by Islam et al. [1]. Differences in split, preprocessing, software version, or undocumented tuning may explain part of the gap. We did not tune toward the published number. The enrolment-time UCI benchmark also cannot be compared directly with full-feature UCI studies because it deliberately removes semester outcomes.

Wu et al. [6] reported ROC-AUC 0.794 and PR-AUC 0.664 for a calibrated 28-day model, and Shaikhanova et al. [7] reported ROC-AUC 0.789 and AP 0.722 at day 14. These studies use different targets and split units. Wu et al. predict the final Withdrawn label and group by presentation; Shaikhanova et al. exclude early withdrawals but combine later failure and withdrawal as risk. Our lower result predicts future withdrawal only and separates student identities. Da Silva et al. [16] are methodologically closer because they use valid event dates in survival models. A direct performance claim requires rerunning each method under one shared cohort, target, cutoff, and split.

5.3 Contribution and Novelty Boundary

The project does not claim a new machine-learning algorithm. Its main contribution is a focused evaluation contrast. We combine a landmark outcome, group-separated validation, a controlled same-record temporal contrast, cluster-level uncertainty, calibration, and a later presentation holdout. The controlled contrast explains why additional history helps, while the operational analysis shows why waiting does not necessarily produce a better service. Presenting both avoids a conclusion that would be easy to draw from either analysis alone. Because recent work already covers survival benchmarks, calibration, capacity, and fairness [6, 16], this is an incremental methodological contribution rather than a new state-of-the-art framework.

5.4 Model Complexity

XGBoost gave a modest advantage over logistic regression on OULAD, especially on the later-presentation holdout. On UCI, the two models had almost identical ROC-AUC and Brier score. This supports a practical rule: use the simpler model when its local validation is comparable, and accept XGBoost only when the improvement is large enough to justify explanation, monitoring, and maintenance work.

6. Practical Implications

The appropriate use is a ranked work list for trained advisors. The system should display a calibrated score, the prediction date, recent assessment and engagement evidence, and any data-quality warning. An advisor should verify context before deciding whether to contact a student. The score should not change grades, restrict enrolment, remove financial support, or initiate another adverse decision.

An illustrative rule that reserves review capacity for about the highest-risk 15% of validation cases produced 819 day-35 test alerts, including 329 true and 491 false alerts. This tradeoff shows why operational capacity and intervention quality matter as much as ranking performance. A prospective pilot should measure timely review, contact, support uptake, persistence, student experience, and advisor workload.

Table 12. Minimum controls for a prospective pilot

Area	Required control
Temporal integrity	Create features at an approved cutoff and block later events
Human oversight	Advisor reviews context before any student contact
Capacity	Set alert volume from available service resources
Monitoring	Track AP, calibration, errors, drift, and subgroup uncertainty
Governance	Limit access, log use, define retention, and allow correction

7. Limitations, Ethics, and Future Work

The study is retrospective. OULAD describes UK distance learning in 2013-2014, and the UCI data come from one Portuguese institution. The 2014J analysis is a later-presentation holdout, not external validation at another institution. Withdrawal reasons are absent, a small number of withdrawn OULAD records lack dates, and the UCI Enrolled class remains unresolved when grouped as non-dropout. The small tuning grid did not include survival models or sequence models. Thresholds were not selected from measured intervention costs, and no field experiment tested whether an alert improves persistence. Recent OULAD studies already cover survival benchmarks and capacity-linked fairness auditing [6, 16]. Our evaluation did not reproduce the fairness analysis, so the manuscript should be treated as a working paper rather than a submission-ready claim of major novelty.

Student information is sensitive even after de-identification. A local system would require purpose limitation, restricted access, audit logs, retention rules, and subgroup evaluation. Demographic and financial variables may encode structural inequalities. SHAP values describe model use of predictors and should not be treated as causal evidence or a prescription for individual students.

Future work should pre-register a local landmark cohort and compare logistic regression, XGBoost, and a discrete-time survival model under the same predictors and student groups. A later term or second institution should remain untouched for external testing. If the institution proceeds to outreach, a randomized or phased pilot is needed to estimate intervention effects.

8. Conclusion

This project began as a comparison of common dropout classifiers and explainability tools. The final audit showed that a more important problem was how the prediction question had been defined. Once we removed already-withdrawn cases and separated students across data partitions, OULAD performance became more modest. The resulting estimates are less impressive than the preliminary scores, but they are more useful for deciding whether a real pilot is justified.

For the same day-75 cohort, additional history from day 35 to day 75 improved ROC-AUC by 0.045 and AP by 0.041. For changing operational cohorts, ROC-AUC stayed near 0.717 while AP declined. The later 2014J holdout reached 0.678 ROC-AUC. Together, these results support a cautious advisor decision aid with local calibration and monitoring. They do not support autonomous action or a claim of ready-to-deploy external validity.

Final recommendation: proceed only to a small, human-reviewed prospective pilot. Treat the current model as research evidence and a reproducible starting point, not a finished institutional product.

Declarations

Data Availability

The UCI Predict Students' Dropout and Academic Success dataset and OULAD are public datasets cited in References [3] and [4]. No new student data were collected.

Code Availability

Analysis code, result tables, figures, and the capstone PDF are available in the public project repository. The main reproducible analysis is `final_report/final_analysis.py`.

Ethics Statement

This study used public, de-identified secondary data and did not contact students or make academic decisions.

Competing Interests

The authors declare no competing interests.

Author Contributions

Both authors contributed to project design, analysis, interpretation, and manuscript preparation. Contribution wording should be converted to the target journal's required CRediT format before submission.

References

- [1] M. M. Islam, F. H. Sojib, M. F. H. Mihad, M. Hasan, and M. Rahman, "The integration of explainable AI in Educational Data Mining for student academic performance prediction and support system," *Telematics and Informatics Reports*, vol. 18, Art. no. 100 203, 2025. doi: 10.1016/j.teler.2025.100203.
- [2] V. Realinho, J. Machado, L. Baptista, and M. V. Martins, "Predicting Student Dropout and Academic Success," *Data*, vol. 7, no. 11, Art. no. 146, 2022. doi: 10.3390/data7 110 146.
- [3] V. Realinho, M. Vieira Martins, J. Machado, and L. Baptista, "Predict Students' Dropout and Academic Success," *UCI Machine Learning Repository*, 2021. doi: 10.24432/C5MC89.
- [4] J. Kuzilek, M. Hlosta, and Z. Zdrahal, "Open University Learning Analytics dataset," *Scientific Data*, vol. 4, Art. no. 170 171, 2017. doi: 10.1038/sdata.2017.171.
- [5] B. Carballo-Mendivil, A. Arellano-Gonzalez, N. J. Rios-Vazquez, and M. del P. Lizardi-Duarte, "Predicting Student Dropout from Day One: XGBoost-Based Early Warning System Using Pre-Enrollment Data," *Applied Sciences*, vol. 15, no. 16, Art. no. 9202, 2025. doi: 10.3390/app15 169 202.
- [6] S. Wu, J. Duan, and M. Luo, "Calibrated early-warning models with fairness auditing and selective prediction for course withdrawal risk: Evidence from OULAD," *PLOS ONE*, vol. 21, no. 7, e0352867, 2026. doi: 10.1371/journal.pone.0352867.
- [7] A. Shaikhanova, O. Kuznetsov, K. Iklassova, A. Tokkuliyeve, and L. Sugurova, "Interpretable Predictive Modeling for Educational Equity: A Workload-Aware Decision Support System for Early Identification of At-Risk Students," *Big Data and Cognitive Computing*, vol. 9, no. 11, Art. no. 297, 2025. doi: 10.3390/bdcc9 110 297.
- [8] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785-794. doi: 10.1145/2 939 672.2939785.
- [9] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [10] G. S. Collins et al., "TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, e078 378, 2024. doi: 10.1136/bmj-2023-078 378.
- [11] K. G. M. Moons et al., "PROBAST+AI: an updated quality, risk of bias, and applicability assessment tool for prediction models using regression or artificial intelligence methods," *BMJ*, vol. 388, e082 505, 2025. doi: 10.1136/bmj-2024-082 505.
- [12] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1, 2023. doi: 10.6028/NIST.AI100-1.
- [13] C. Romero and S. Ventura, "Educational data mining and learning analytics: An updated survey," *WIREs Data Mining and Knowledge Discovery*, vol. 10, no. 3, e1355, 2020. doi: 10.1002/widm.1355.
- [14] F. Qiu et al., "Predicting students' performance in e-learning using learning process and behaviour data," *Scientific Reports*, vol. 12, Art. no. 453, 2022. doi: 10.1038/s41 598-021-03 867-8.
- [15] S. Plak, I. Cornelisz, M. Meeter, and C. van Klaveren, "Early warning systems for more effective student counselling in higher education: Evidence from a Dutch field experiment," *Higher Education Quarterly*, vol. 76, no. 1, pp. 131-152, 2022. doi: 10.1111/hequ.12298.
- [16] R. da Silva, J. Eicher, and G. Longo, "Temporal Dropout Risk in Learning Analytics: A Harmonized Survival Benchmark Across Dynamic and Early-Window Representations," *arXiv:2604.08870*, 2026. doi: 10.48550/arXiv.2604.08870.